

Thomas Lepère

Master student in Mechanical Engineering

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[thomas-toignons.github.io/Lepere_Portfolio](https://github.com/thomas-toignons/Lepere_Portfolio)

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in [Thomas Lepère](#)



Summary | Internship in fluid mechanics simulation or correlation

Mechanical Engineering MSc student specializing in fluid mechanics, with hands-on experience in wind-tunnel testing, CFD simulation and validation, and data analysis. Seeking a 3- to 6-month internship in these fields starting in September 2026.

EDUCATION

M. Sc in Mechanical Engineering

2023 - present
EPFL, Lausanne

- 2024 - present : Fluid Mechanics Specialization
- 2023 - 2024 : HES gateway (2nd and 3rd year Bachelor courses)

B. Sc in Mechanical Engineering

2020 - 2023
HEPIA, Genève

- Fluid Mechanics and Energy specialization
- Bachelor Thesis on CFD correlation of a foiling catamaran

EXPERIENCE

Head of CFD validation at **EPFL** **Racing team**

2025 - 2026
EPFL, Lausanne

- Designed, manufactured, and wind-tunnel tested a reduced-scale race car model
- Contributed to CFD methodology validation through comparison with experimental data

Laboratory assistant at **CRYOS**

2025
EPFL, Sion

- Compared the performance of two machine-learning models for weather prediction using Python
- Automated data processing, evaluated prediction accuracy, and produced structured result summaries

Teaching Assistant

2025
EPFL, Lausanne

- Supported students during finite element modelling sessions using Abaqus

Sailing Instructor

2016 - 2025
CNV, Versoix

- Supervised groups of children aged 8 to 18 during weekends and summer camps
- Ensured safety, planning, and skill progression

Trainee in a school workshop

2019 - 2020
HEPIA, Genève

- Worked on lathes and milling machines
- Designed a custom MS Access database to optimize the manufacturing process

EXTRA-PROFESSIONAL ACTIVITIES

Head of electricity at **Balélec**

2025 - 2026
EPFL, Lausanne

- Responsible for electricity supply in the logistic department
- Multidisciplinary project management
- Recruited and supervised team members
- Electrical safety training (ElectroSuisse)
- Rooftop operator training

SKILLS

Technical: Abaqus, ANSYS Fluent, Creo Parametric, LabView, Matlab, MS Access, MS Excel, Python, 3DEXperience, 3D printing

Languages: French (Native), English (Fluent), Spanish (A2)

PROJECTS

- Wind tunnel study of a racing boat's nets for CFD modeling
- CFD study on the effect of slot width on two-element wingsails prediction
- Machine learning model to assess wind turbine performance
- Simulation of particle-laden thermally buoyant horizontal jets

AWARDS

- Winner of the Translémanique en Solitaire (24 hours regatta on the Geneva Lake), 2025
- Academic Excellence Award for the best mean over the Bachelor, HEPIA, 2023
- Most original solution for a fire extinguishing robot, HEPIA, 2020

REFEREES

Fabien Breda, HEPIA

@ fabien.breda@hesge.ch

Michael Lehning, EPFL

@ michael.lehning@epfl.ch

ABOUT ME

Interests: Sailing (high-level competitor from 2017 to 2019), F1, climbing, skiing, cycling, video games, and board games

25 years old, born 02/05/2001

Residence permit B

License B